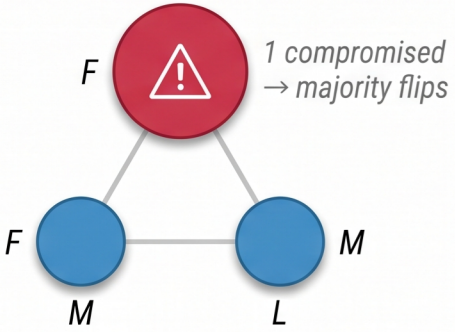
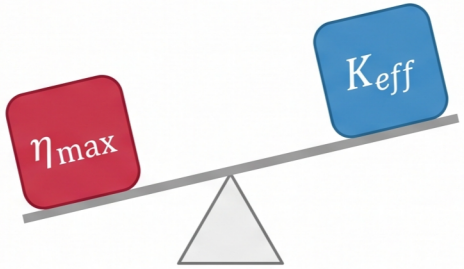
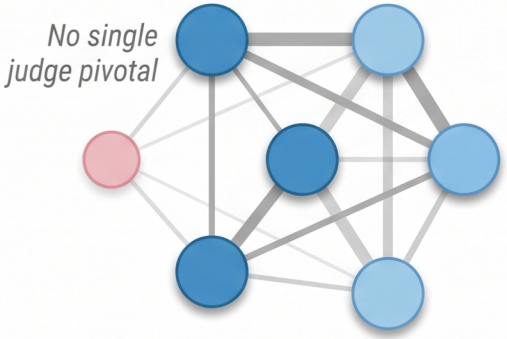
(a) Small Panel (*K*=3)



(b) Predictor Crossover



(c) Large Panel (*K*≥5)



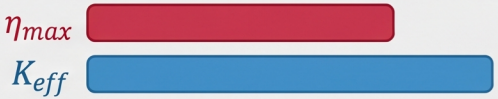
n = 455 panels

η_{max} β = 0.264 [0.158, 0.369]

K_{eff} β = 0.349

Significance: η_{max} 10/12, K_{eff} 12/12 (OLS)

R^2 = 0.327 (majority vote)

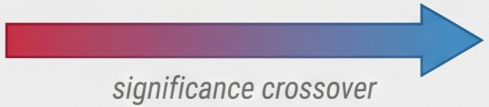


$K^* \in (3, 5)$

η_{max} CI crosses zero at *K*≥5

K_{eff} positive 12/12 conditions

MDE: 0.14 → 0.048

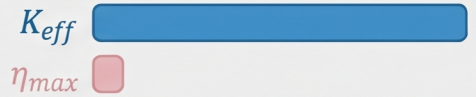


n = 3,003–6,435 panels

K_{eff} β = 0.437 [0.318, 0.553]

η_{max} β = 0.084 [−0.083, 0.251]

Significance: K_{eff} 7/12, η_{max} 2/12 (bootstrap)



Attack type: *targeted vs shared-bias → no differential effect (KW p = 0.200)*

Aggregation sensitivity: *Bayesian R^2 = 0.09 neutralizes diversity signal*